

ZHONGWEI ZHANG (Zachary)

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PERSONAL STATEMENT

Master of Information Technology student at the University of Auckland with previous experience in embedded software, industrial control systems and HMI integration. Interested in software engineering, full-stack web applications and systems/embedded development. Built team projects with React, Next.js, TypeScript, Svelte, Node.js, SQL, PostgreSQL and SQLite, bringing practical engineering, systematic debugging and collaborative development skills.

KEY COMPETENCIES

- Developed full-stack web applications using Svelte, Node.js, Next.js, React, SQL, PostgreSQL and SQLite.
- Designed responsive interfaces and interactive browser experiences using TypeScript, Tailwind CSS, HTML and CSS.
- Implemented REST APIs, authentication, password hashing, persistence, search/filtering and file-upload workflows.
- Diagnosed UI, state, audio, persistence and controller-related issues using systematic debugging and automated testing.
- Collaborated in 4-6 person teams using GitHub branches, pull requests, code review and iterative integration.

EDUCATION

Master of Information Technology, University of Auckland Jul 2025 - Present

- Achieved A+ grades in CS718 and CS719; relevant coursework includes CS718, CS719 and CS732.

Bachelor of Mechanical Engineering, Xi'an Jiaotong University 2018 - 2022

RELEVANT WORK EXPERIENCE

Software / Embedded Systems Engineer, State Grid Corporation of China 2022 - 2025

China

- Monitored and maintained PID controller systems in industrial environments, supporting stable operations and timely fault response.
- Programmed embedded C routines and diagnosed controller-related software, reliability and integration issues.
- Integrated hardware and HMI functionality while documenting findings and coordinating solutions with engineering stakeholders.

SELECTED ICT PROJECTS

Last Answer - Browser-Based Fantasy Quiz RPG | CS732 2026

Team 8-Bit

React / Next.js / TypeScript / Tailwind CSS / Zustand / Vercel

- Designed the core quiz-battle UI architecture, including combat stages, timers, status panels, answer grids, logs, overlays and outcome screens.
- Engineered turn-based interactions, item/support logic, question randomisation, post-battle state management and layered scene audio.
- Collaborated through GitHub feature development, pull-request integration and project-wide debugging; deployed the application on Vercel.

Full-Stack Blog Website | University of Auckland Team Project 2025

Svelte / Node.js / SQLite / REST APIs / GitHub

- Collaborated in a four-person team to build a responsive blog platform and integrate front-end and back-end features.
- Developed login/registration APIs, username validation, password-strength feedback, secure hashing, article search/filtering and image uploads.

Currency Exchange Web Application | University of Auckland Team Project 2026

Svelte / JavaScript / Chart.js / localStorage / Vercel

- Built interactive exchange-rate visualisations and user-defined threshold alerts with browser persistence, then tested and integrated features with the team.

TECHNICAL SKILLS

Programming

Frameworks & Tools

Databases & Areas

JavaScript, TypeScript, HTML, CSS, SQL, Java, C
Next.js, React, Node.js, Svelte, Tailwind CSS, Zustand, Chart.js, Git,
GitHub, Vitest, Playwright, Vercel
PostgreSQL, SQLite; REST APIs, authentication, responsive UI,
state management, testing, embedded systems, HMI